

2024 Calculus II Additional Exercise (14)

1. Exercise 37 in page 55 from Calculus Supplement Homework.
2. Exercise 17 in page 54 from Calculus Supplement Homework.(Hint: Using Stokes's Theorem)
3. Exercise 24 in page 54 from Calculus Supplement Homework.
4. Exercise 28 in page 55 from Calculus Supplement Homework.(Hint: Using Stokes's Theorem)
5. Exercise 35 in page 55 from Calculus Supplement Homework.

补充作业 14

Q1. P55 Ex 37

37. Let S be the part of the plane $x + y + z = 1$ in the first octant, compute

$$\iint_S yz \, d\sigma.$$

卦限

Sol.: 参数化 $\begin{cases} x = u \\ y = v \\ z = 1 - u - v \end{cases}$

$$\vec{r}_u = \frac{\partial \vec{r}}{\partial u} = \vec{i} - \vec{k}$$

$$\vec{r}_v = \frac{\partial \vec{r}}{\partial v} = \vec{j} - \vec{k}$$

$$\Rightarrow |\vec{r}_u \times \vec{r}_v| = \left| \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} \right| = |\vec{i} + \vec{j} + \vec{k}| = \sqrt{3}$$

$$\iint_S yz \, d\sigma = \iint_R y(u, v) z(u, v) |\vec{r}_u \times \vec{r}_v| \, du \, dv$$

$$= \iint_R v(1-u-v) \sqrt{3} \, du \, dv$$

$$= \sqrt{3} \int_0^1 \left(\int_0^{1-v} v - v^2 - vu \, du \right) dv$$

$$= \sqrt{3} \int_0^1 (v - v^2)(1-v) - \frac{v}{2}(1-v)^2 \, dv$$

$$= \frac{\sqrt{3}}{2} \int_0^1 v - 2v^2 + v^3 \, dv$$

$$= \frac{\sqrt{3}}{2} \left(\frac{v^2}{2} - \frac{2}{3}v^3 + \frac{v^4}{4} \right) \Big|_0^1$$

$$= \frac{\sqrt{3}}{24}$$

$$\begin{aligned} & \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) \\ &= \frac{6-8+3}{12} \\ &= \frac{1}{12} \end{aligned}$$

Q2 PS4 Ex17

P Q R

17. Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = (y^2 - z^2, 2z^2 - x^2, 3x^2 - y^2)$ and C is the intersection of plane $x + y + z = 2$ and cylinder $|x| + |y| = 1$. In the counter-clockwise direction if seen from the positive direction of z .

Sol: Use Stokes Thm for closed curve C .

$$\nabla \times \mathbf{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

$$= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) \vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) \vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \vec{k}$$

$$= (-2y - 4z) \vec{i} + (-2z - 6x) \vec{j} + (-2x - 2y) \vec{k}$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \vec{n} d\sigma$$

参数化曲面 S : $\begin{cases} x = u \\ y = v \\ z = 2 - u - v \end{cases}$

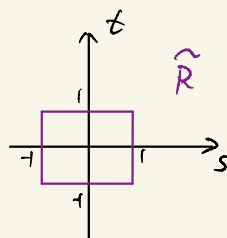
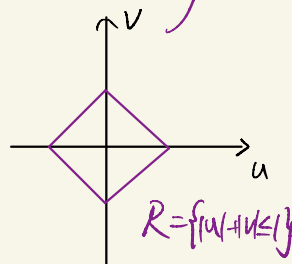
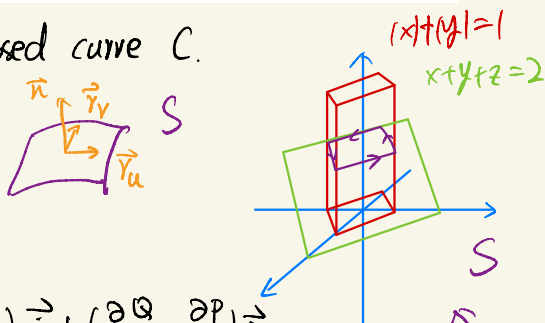
$$\begin{cases} \vec{r}_u = \frac{\partial \vec{r}}{\partial u} = \vec{i} - \vec{k} \\ \vec{r}_v = \frac{\partial \vec{r}}{\partial v} = \vec{j} - \vec{k} \end{cases} \Rightarrow \vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} = \frac{\vec{i} + \vec{j} + \vec{k}}{\sqrt{3}}$$

$$\therefore \iint_S (\nabla \times \mathbf{F}) \cdot \vec{n} d\sigma = \frac{1}{\sqrt{3}} \iint_S (-2y - 4z - 2z - 6x - 2x - 2y) d\sigma$$

$$= \frac{1}{\sqrt{3}} \iint_R (-8u - 4v - 6(2 - u - v)) |\vec{r}_u \times \vec{r}_v| dA$$

$$= \iint_R (-12 - 2u + 2v) du dv$$

换元: $\begin{cases} s = u + v \\ t = u - v \end{cases} \Rightarrow \begin{cases} u = \frac{1}{2}(s + t) \\ v = \frac{1}{2}(s - t) \end{cases} \Rightarrow \frac{\partial(u, v)}{\partial(s, t)} = \frac{1}{2}$



$$\text{原式} = \iint_{\widehat{R}} (-12 + 2t) \cdot \frac{1}{2} ds dt$$

$$= \int_{-1}^1 \int_{-1}^1 -6 + t dt ds$$

$$= \int_{-1}^1 -12 ds$$

$$= -24$$

17

Q3 P55 Ex24

24. Use the Stokes' Theorem to calculate the circulation of the field $\mathbf{F} = (y, -xz, x^2)$ around the curve C , which is the boundary of the triangle cut from the plane $x + y + z = 1$ by the first octant, counterclockwise when views from above.

Sol. $\vec{n}_s = \frac{1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$

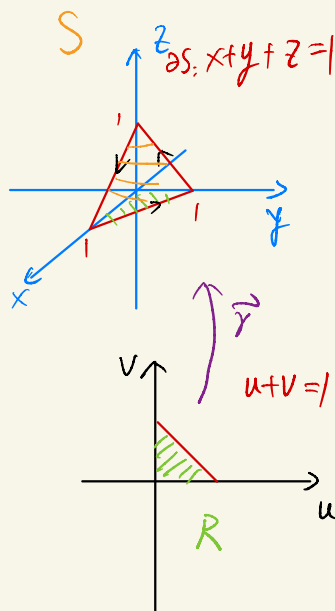
$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & -xz & x^2 \end{vmatrix} = x\vec{i} - 2x\vec{j} + (-z-1)\vec{k}$$

By Stokes' Thm:

$$\oint_{\partial S} \vec{F} d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n}_s d\sigma$$

$$= \frac{1}{\sqrt{3}} \iint_S x - 2x - z - 1 d\sigma$$

原式 $= \frac{1}{\sqrt{3}} \iint_S -x - z - 1 d\sigma$



曲面参数化:

$$\vec{r}(u, v) = (x(u, v), y(u, v), z(u, v)) \quad \begin{cases} x = u \\ y = v \\ z = 1 - u - v \end{cases} \quad (u, v) \in R$$

$$\vec{r}'_u = \vec{i} - \vec{k} \quad \vec{r}'_v = \vec{j} - \vec{k} \quad \Rightarrow \quad \vec{n} = \frac{\vec{r}'_u \times \vec{r}'_v}{|\vec{r}'_u \times \vec{r}'_v|} = \frac{1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$$

□

$$\begin{aligned}
\therefore \text{原式} &= \frac{1}{\sqrt{3}} \iint_R (-u - (1-u-v) - 1) |\vec{r}'_u \times \vec{r}'_v| \, du \, dv \\
&= \iint_R v - 2 \, du \, dv \\
&= \int_0^1 \int_0^{1-u} v - 2 \, dv \, du \\
&= \int_0^1 \frac{(1-u)^2}{2} - 2(1-u) \, du \\
&= \frac{1}{2} \int_0^1 u^2 - 2u + 1 - 4 + 4u \, du \\
&= \frac{1}{2} \int_0^1 u^2 + 2u - 3 \, du \\
&= \frac{1}{2} \left(\frac{1}{3} + 1 - 3 \right) \\
&= -\frac{5}{6}
\end{aligned}$$

12

Q4. PSS Ex28

28. Let L be the intersection of $y + z = 2$ and $x^2 + y^2 = 1$ in the counterclockwise direction if view from the positive part of z -axis. Compute

$$\oint_L -y^2 dx + x dy + z^2 dz.$$

Sol: Use Stokes' Thm.

Let vector field $\vec{F} = -y^2 \vec{i} + x \vec{j} + z^2 \vec{k}$

by Stokes' Thm:

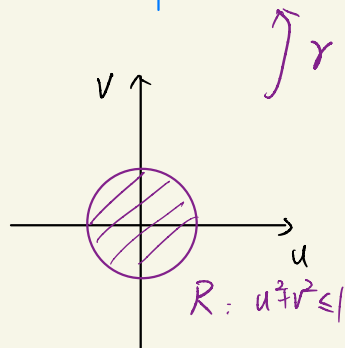
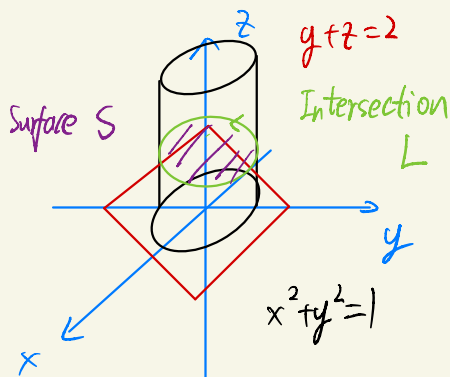
$$\begin{aligned} & \oint_L -y^2 dx + x dy + z^2 dz \\ &= \oint_L \vec{F} \cdot d\vec{r} \end{aligned}$$

$$= \iint_S (\nabla \times \vec{F}) \cdot \vec{n} d\sigma$$

计算: $\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & x & z^2 \end{vmatrix} = (1+2y) \vec{k}$

参数化曲面 S : $\begin{cases} x = u \\ y = v \\ z = 2 - v \end{cases} \Rightarrow \begin{cases} \vec{r}_u = [\vec{i} \ \vec{j} \ \vec{k}] [1 \ 0 \ 0]^T \\ \vec{r}_v = [\vec{i} \ \vec{j} \ \vec{k}] [0 \ 1 \ -1]^T \end{cases} \Rightarrow \vec{r}_u \times \vec{r}_v = \vec{j} + \vec{k}$

$$\begin{aligned} \therefore \iint_S (\nabla \times \vec{F}) \cdot \vec{n} d\sigma &= \iint_S (\nabla \times \vec{F}) \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} d\sigma \\ &= \iint_R (1+2y) \cdot \frac{|\vec{r}_u \times \vec{r}_v|}{|\vec{r}_u \times \vec{r}_v|} dA \\ &= \int_{-1}^1 \left(\int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} (1+2v) dv \right) du \\ &= \pi \end{aligned}$$



Q5 P55 Ex 35

35. Parametrize the following surfaces and derive the factor $|r_u \times r_v|$.

(1) the sphere centered at the origin of radius R ;

(2) a circular cylinder $y^2 + z^2 = a^2$;

(3) a cone $x = 2\sqrt{y^2 + z^2}$;

(4) a paraboloid $z = 16 - x^2 - y^2$.

Sol. (1)

$$\begin{cases} x = u \\ y = v \\ z = \sqrt{a^2 - u^2 - v^2} \end{cases}$$

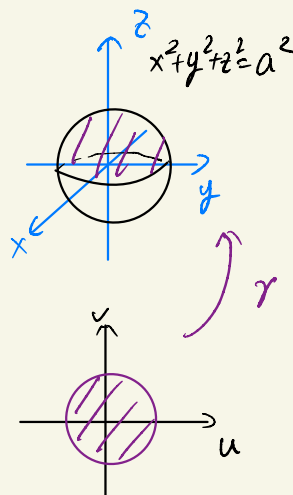
$$\vec{r}'_u = \left(1, 0, \frac{-u}{\sqrt{a^2 - u^2 - v^2}} \right)$$

$$\vec{r}'_v = \left(0, 1, \frac{-v}{\sqrt{a^2 - u^2 - v^2}} \right)$$

$$\vec{r}'_u \times \vec{r}'_v = \left(\frac{u}{\sqrt{a^2 - u^2 - v^2}}, \frac{v}{\sqrt{a^2 - u^2 - v^2}}, 1 \right)$$

$$|\vec{r}'_u \times \vec{r}'_v| = \frac{\sqrt{u^2 + v^2 + a^2 - u^2 - v^2}}{\sqrt{a^2 - u^2 - v^2}} = \frac{a}{\sqrt{a^2 - u^2 - v^2}}$$

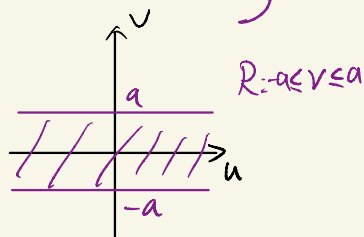
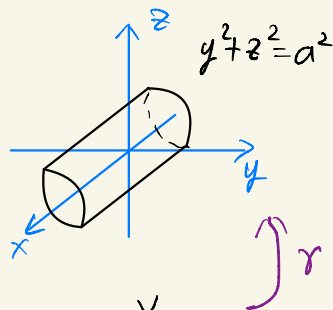
类似地也可计算下半球壳



(2) $\begin{cases} x = u \\ y = v \\ z = \sqrt{a^2 - v^2} \end{cases} \quad ((u, v) \in \mathbb{R})$ 上半圆柱参数化.

$$\vec{r}'_u = (1, 0, 0)$$

$$\vec{r}'_v = \left(0, 1, \frac{-v}{\sqrt{a^2 - v^2}} \right)$$



$$\vec{r}'_u \times \vec{r}'_v = (0, \frac{v}{\sqrt{a^2 - v^2}}, 1)$$

$$|\vec{r}'_u \times \vec{r}'_v| = \frac{a}{\sqrt{a^2 - v^2}}$$

类似可计算下半圆柱

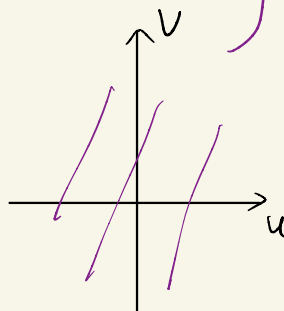
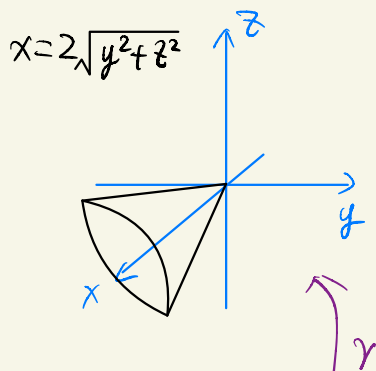
$$(3) \begin{cases} x = 2\sqrt{u^2 + v^2} \\ y = u \\ z = v \end{cases}$$

$$\vec{r}'_u = (\frac{2u}{\sqrt{u^2 + v^2}}, 1, 0)$$

$$\vec{r}'_v = (\frac{2v}{\sqrt{u^2 + v^2}}, 0, 1)$$

$$\vec{r}'_u \times \vec{r}'_v = (1, \frac{-2u}{\sqrt{u^2 + v^2}}, \frac{-2v}{\sqrt{u^2 + v^2}})$$

$$|\vec{r}'_u \times \vec{r}'_v| = \sqrt{5}$$



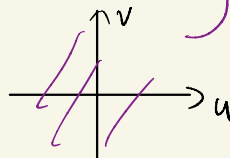
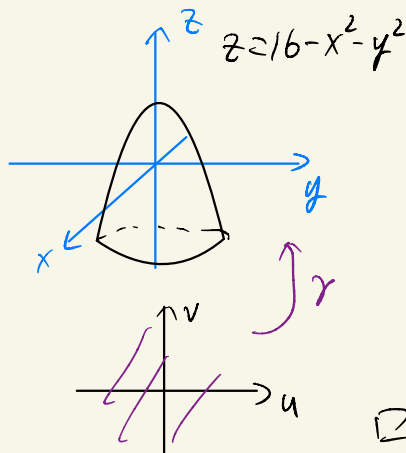
$$(4) \begin{cases} x = u \\ y = v \\ z = 16 - u^2 - v^2 \end{cases}$$

$$\vec{r}'_u = (1, 0, -2u)$$

$$\vec{r}'_v = (0, 1, -2v)$$

$$\vec{r}'_u \times \vec{r}'_v = (2u, 2v, 1)$$

$$|\vec{r}'_u \times \vec{r}'_v| = \sqrt{1 + 4u^2 + 4v^2}$$



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